Unit: 6 Unit 5: Linear Equations
Topic: Scatter Plot and The Best Fit Line

Name: ____________
Date: __________

Homework (Jalapeno)

Q1. In ancient Egyptian architecture, the height h of a pyramid is related to its base perimeter p. Which type of correlation is observed in the scatter plot?
(A) Positive
(B) Negative
(C) None
(D) Cannot be determined
(E) Not enough information

Q2. A timeline of the average temperature T in degrees Celsius and the number of sunspots s observed over the years is plotted. If the line of best fit has a slope of -0.5, what does this indicate?
(A) For every increase in sunspots, temperature decreases by 0.5^circC
(B) For every decrease in sunspots, temperature increases by 0.5^circC
(C) For every increase in sunspots, temperature increases by 0.5^circC
(D) For every decrease in sunspots, temperature decreases by 0.5^circC
(E) No relationship between sunspots and temperature

Q3. The legacy of ancient Greek philosophers is studied in relation to their age a and the number of written works w. Which of the following is a characteristic of a positive correlation?
(A) As age increases, number of written works decreases
(B) As age decreases, number of written works increases
(C) As age increases, number of written works increases
(D) No relationship between age and number of written works
(E) The data is inconsistent

Q4. Artifacts from an ancient civilization show a relationship between the weight w of a statue and its height h. If the line of best fit is $h = 2w + 1$, what is the predicted height of a statue that weighs 3 kg?
(A) 5
(B) 6
(C) 7
(D) 8
(E) 9

Q5. A historical analysis of the population p of a city and the number of diseases d reported over time shows a correlation. Which of the following is a characteristic of a negative correlation?
(A) As population increases, number of diseases increases
(B) As population decreases, number of diseases decreases
(C) As population increases, number of diseases decreases
(D) No relationship between population and number of diseases
(E) The data is inconsistent

Q6. The construction of the Great Pyramid of Giza took approximately 20 years to complete. If the number of workers w and the time t taken to complete a certain portion of the pyramid are related by the equation $t = \frac{1}{2}w + 5$, what is the predicted time taken to complete the portion with 10 workers?
(A) 5 years
(B) 7.5 years
(C) 10 years
(D) 12.5 years
(E) 15 years

Q7. An ancient timeline shows the relationship between the number of wars w fought and the number of years y of peace that follow. If the line of best fit is $y = -2w + 10$, what is the predicted number of years of peace after 2 wars?
(A) 2 years
(B) 4 years
(C) 6 years
(D) 8 years
(E) 10 years

Q8. The ancient Olympic Games had a record of the distance d of the long jump and the time t taken to complete the jump. If the correlation between d and t is positive, what can be concluded?
(A) As distance increases, time decreases
(B) As distance decreases, time increases
(C) As distance increases, time increases
(D) No relationship between distance and time
(E) The data is inconsistent

Open Ended Questions (FRQ)

Q9. A scatter plot shows the relationship between the height h of a volcano and the number of eruptions e over a period of time. If the line of best fit is $h = 0.5e + 2$, predict the height of the volcano after 5 eruptions and explain your reasoning.

Q10. The legacy of an ancient civilization is studied in relation to the number of artifacts a discovered and the age y of the artifacts. If a scatter plot shows a negative correlation between a and y, draw a possible line of best fit and explain what this indicates about the relationship between the number of artifacts and their age.

Chef's Tips

- Tip 1: Ensure students understand the concept of correlation and how to interpret it.
- Tip 2: Emphasize the importance of using the line of best fit to make predictions.
- Tip 3: Encourage students to use real-world examples to illustrate their understanding of the concepts.